AI MOVIE RECOMMENDATION SYSTEM

Project Report

1.Introduction:

The AI Movie Recommendation System is a project that suggests movies to a user based on the movie they choose. The main idea is to find movies that are similar in terms of their genres, keywords, and description. I developed this project using Python and machine learning libraries.

2. Methodology:

First, I loaded a movie dataset using Pandas. The dataset contains information about different movies.

2.1 Data Selection

I selected the important columns needed for the recommendation system, including movie title, genres, keywords, and overview.

2.2 Creating Movie Tags

I combined the genres, keywords, and movie overview into one text field called tags. This gave the system useful information about each movie in one place.

2.3 Text Vectorization

I used CountVectorizer from Scikit-learn to convert the movie tags into numerical vectors. This allows the computer to compare the movie information.

2.4 Cosine Similarity

I used cosine similarity to calculate how similar the movies are to each other. Movies with more similar information get a higher similarity score.

2.5 Recommendation Function

Finally, I created a recommendation function. When a movie title is entered, the system finds that movie and displays five movies that are most similar to it.

3. Technologies Used:

Python, Pandas, Scikit-learn, CountVectorizer, and Cosine Similarity were used to build this project. The project was developed and tested in Google Colab using a Jupyter Notebook.

4. Results:

The system successfully generated movie recommendations for different input movies. I tested the system with movies such as Avatar, Inception, The Dark Knight, The Matrix, Jurassic Park, and Titanic. For each movie, the system displayed five movies that were considered similar.

5. Insights:

While working on this project, I learned how movie information can be used to build a simple recommendation system. I also learned how text data can be converted into numerical values and how cosine similarity can be used to compare items. I noticed that some movies can appear in more than one recommendation list because they have similar features.

6. Conclusion:

The AI Movie Recommendation System successfully recommends movies based on their similarity to a movie selected by the user. This project helped me understand data preprocessing, text vectorization, similarity calculation, and the basic idea behind recommendation systems. It also gave me practical experience using Python and Scikit-learn.

7. Future Improvements:

In the future, the system could be improved by using user ratings, watch history, or more movie features. A graphical user interface could also be added to make the system easier to use.